

$$\dot{x}(t) = A\sigma(t) x(t) + b\sigma(t)$$

$$x(t) = \begin{bmatrix} x(t) \\ 1 \end{bmatrix}$$

$$\dot{x}(t) = \begin{bmatrix} \dot{x}(t) \\ \frac{d}{dt}(1) \end{bmatrix} = \begin{bmatrix} A x(t) + b\sigma(t) \\ 0 \end{bmatrix}$$

$$\dot{x}(t) = \underbrace{\begin{bmatrix} A\sigma(t) & b\sigma(t) \\ 0 & 0 \end{bmatrix}}_{\tilde{A}\sigma(t)} \begin{bmatrix} x(t) \\ 1 \end{bmatrix}$$

Equação dinâmica com estados aumentados.

$$\underline{\underline{\dot{x}(t) = \tilde{A}\sigma(t) x(t)}}$$

$$x(t_N) = e^{\tilde{A}_{N-1}\Delta t_N} e^{\tilde{A}_{N-2}\Delta t_{N-1}} \dots e^{\tilde{A}_0\Delta t_1} x(t_0)$$

$$F_{i+1} = \tilde{A}\sigma(t_i)$$

$$x(t_N) = e^{F_N\Delta t_N} e^{F_{N-1}\Delta t_{N-1}} \dots e^{F_1\Delta t_1} x(t_0)$$

Da mesma forma

$$\bar{x}(t_N) = e^{F_N\Delta \bar{t}_N} e^{F_{N-1}\Delta \bar{t}_{N-1}} \dots e^{F_1\Delta \bar{t}_1} \bar{x}(t_0)$$

$$e(t) = x(t) - \bar{x}(t)$$

$$\Delta t_i = \Delta \bar{t}_i + \delta t_i$$

$$e(t_N) = e^{f_N \Delta t_N} e^{f_{N-1} \Delta t_{N-1}} \dots e^{f_1 \Delta t_1} x(t_0) - e^{f_N \Delta \bar{t}_N} e^{f_{N-1} \Delta \bar{t}_{N-1}} \dots e^{f_1 \Delta \bar{t}_1} \bar{x}(t_0)$$

$$x(t_0) = \bar{x}(t_0) + e(t_0)$$

$$= e^{f_N(\Delta \bar{t}_N + \delta t_N)} e^{f_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})} \dots e^{f_1(\Delta \bar{t}_1 + \delta t_1)} (\bar{x}(t_0) + e(t_0)) - e^{f_N \Delta \bar{t}_N} e^{f_{N-1} \Delta \bar{t}_{N-1}} \dots e^{f_1 \Delta \bar{t}_1} \bar{x}(t_0)$$

Usando a aproximação de primeira ordem:

$$e^{At_b} \doteq e^{At_a} + A e^{At_a} (t_b - t_a)$$

$$e^{A \delta t} \doteq I + A \delta t$$

$$e^{F(\Delta t + \delta t)} = e^{F \Delta t} + A e^{F \Delta t} \cdot \delta t \quad ?$$

$$e(t_N) = e^{f_N(\Delta \bar{t}_N + \delta t_N)} e^{f_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})} \dots e^{f_1(\Delta \bar{t}_1 + \delta t_1)} (\bar{x}(t_0) + e(t_0)) - e^{f_N \Delta \bar{t}_N} e^{f_{N-1} \Delta \bar{t}_{N-1}} \dots e^{f_1 \Delta \bar{t}_1} \bar{x}(t_0)$$

$$= (e^{f_N \Delta \bar{t}_N} + f_N e^{f_N \Delta \bar{t}_N} \delta t_N) \cdot (e^{f_{N-1} \Delta \bar{t}_{N-1}} + f_{N-1} e^{f_{N-1} \Delta \bar{t}_{N-1}} \delta t_{N-1}) \dots (e^{f_1 \Delta \bar{t}_1} + f_1 e^{f_1 \Delta \bar{t}_1} \delta t_1) (\bar{x}(t_0) + e(t_0)) - e^{f_N \Delta \bar{t}_N} e^{f_{N-1} \Delta \bar{t}_{N-1}} \dots e^{f_1 \Delta \bar{t}_1} \bar{x}(t_0)$$

$$\phi_N = e^{f_N \Delta \bar{t}_N}$$

$$= (\phi_N + f_N \phi_N \delta t_N) (\phi_{N-1} + f_{N-1} \phi_{N-1} \delta t_{N-1}) \dots (\phi_1 + f_1 \phi_1 \delta t_1) (\bar{x}(t_0) + e(t_0)) - \phi_N \phi_{N-1} \dots \phi_1 \bar{x}(t_0)$$

$$e(t_N) = e^{f_N(\Delta \bar{t}_N + \delta t_N)} e^{f_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})} \dots e^{f_1(\Delta \bar{t}_1 + \delta t_1)} (\bar{x}(t_0) + e(t_0)) +$$

$$e^{f_N \Delta \bar{t}_N} e^{f_{N-1} \Delta \bar{t}_{N-1}} \dots e^{f_1 \Delta \bar{t}_1} \bar{x}(t_0)$$

$$e(t_N) = (e^{f_N \Delta \bar{t}_N} + f_N e^{f_N \Delta \bar{t}_N} \delta t_N) \cdot (e^{f_{N-1} \Delta \bar{t}_{N-1}} + f_{N-1} e^{f_{N-1} \Delta \bar{t}_{N-1}} \delta t_{N-1}) \cdot \dots$$

$$(e^{f_1 \Delta \bar{t}_1} + f_1 e^{f_1 \Delta \bar{t}_1} \delta t_1) (\bar{x}(t_0) + e(t_0))$$

$$e^{f_N \Delta \bar{t}_N} e^{f_{N-1} \Delta \bar{t}_{N-1}} \dots e^{f_1 \Delta \bar{t}_1} \bar{x}(t_0)$$

$$\phi_N = e^{f_N \Delta \bar{t}_N}$$

$$e(t_N) = (\phi_N + f_N \phi_N \delta t_N) (\phi_{N-1} + f_{N-1} \phi_{N-1} \delta t_{N-1}) \dots (\phi_1 + f_1 \phi_1 \delta t_1) (\bar{x}(t_0) + e(t_0))$$

$$- \phi_N \phi_{N-1} \dots \phi_1 \bar{x}(t_0)$$

$$= \phi_N \phi_{N-1} \cancel{\phi_{N-2}} \dots \phi_1 \bar{x}(t_0) +$$

$$\phi_N \phi_{N-1} \phi_{N-2} \dots \phi_1 e(t_0) +$$

$$[f_N \phi_N \phi_{N-1} \phi_{N-2} \dots \phi_1 \delta t_N$$

$$+ f_{N-1} \phi_N \phi_{N-1} \dots \phi_1 \delta t_{N-1}$$

$$+ f_{N-2} \phi_N \phi_{N-1} \dots \phi_1 \delta t_{N-2}$$

⋮

$$+ f_1 \phi_N \phi_{N-1} \dots \phi_1 \delta t_1] (\bar{x}(t_0) + e(t_0))$$

$$- \phi_N \phi_{N-1} \dots \phi_1 \bar{x}(t_0)$$

Desprezando multiplicação de $\delta t * e(t_0)$

$$\begin{aligned}
 e(t_N) \doteq & \phi_N \phi_{N-1} \phi_{N-2} \cdots \phi_1 e(t_0) + \\
 & [F_N \phi_N \phi_{N-1} \phi_{N-2} \cdots \phi_1 \delta t_N \\
 & + F_{N-1} \phi_N \phi_{N-1} \cdots \phi_1 \delta t_{N-1} \\
 & + F_{N-2} \phi_N \phi_{N-1} \cdots \phi_1 \delta t_{N-2} \\
 & \vdots \\
 & + F_1 \phi_N \phi_{N-1} \cdots \phi_1 \delta t_1] \bar{x}(t_0)
 \end{aligned}$$

$$\Phi = \phi_N \phi_{N-1} \cdots \phi_1$$

$$\begin{aligned}
 e(t_N) \doteq & \Phi e(t_0) + \\
 & F_1 \Phi x(t_0) \delta t_1 + \\
 & F_2 \Phi x(t_0) \delta t_2 + \\
 & \vdots \\
 & F_N \Phi x(t_0) \delta t_N
 \end{aligned}$$

$$\Gamma = \begin{bmatrix} F_1 \Phi x(t_0) \\ F_2 \Phi x(t_0) \\ \vdots \\ F_N \Phi x(t_0) \end{bmatrix}^T ; \quad \delta t = \begin{bmatrix} \delta t_1 \\ \delta t_2 \\ \vdots \\ \delta t_N \end{bmatrix}$$

$$e(t_N) = \Phi e(t_0) + \Gamma \delta t$$

$$\phi_n = e^{F_n \delta t_n}$$

Kawa:

$$\phi_N \phi_{N-1} \dots \phi_1 e(t_0)$$

$$+ \begin{bmatrix} F_N \phi_N \phi_{N-1} \dots \phi_1 \delta t_N \\ \phi_N F_{N-1} \phi_{N-1} \dots \phi_1 \delta t_{N-1} \\ \dots \\ \phi_N \phi_{N-1} \dots \phi_2 F_1 \phi_1 \delta t_1 \end{bmatrix} \bar{x}(t_0)$$

$$\phi_N \phi_{N-1} \dots \phi_1 e(t_0)$$

$$+ \begin{bmatrix} F_N \phi_N \phi_{N-1} \dots \phi_1 \bar{x}(t_0) \\ \phi_N F_{N-1} \phi_{N-1} \dots \phi_1 \bar{x}(t_0) \\ \dots \\ \phi_N \phi_{N-1} \dots \phi_2 F_1 \phi_1 \bar{x}(t_0) \end{bmatrix}^T \begin{bmatrix} \delta t_N \\ \delta t_{N-1} \\ \dots \\ \delta t_1 \end{bmatrix}$$